

# Introduction to lattice based cryptography

arnaucube

September 2026, Barcelona

# Motivation

- till now most cryptography relied on hardness of finding prime factors or finding the discrete logarithm, ie. RSA, Elliptic Curves, etc
- Shor's algorithm suggests that factorization can be done in polynomial time (on quantum computers).
  - given an odd composite  $n \in \mathbb{Z}$ , find  $d \in \mathbb{Z}$ , such that  $1 < d < n$  and  $d|n$ .
- even if quantum computers never reach, the threat is there, so we must prepare
- since 2023/2024 Chrome and Firefox have lattice-based cryptography to the browser for TLS
  - hybrid: X25519 + ML-KEM to establish a symmetric encryption key
  - protect against *harvest now, decrypt later* attacks

# Motivation (in Ethereum's ecosystem)

- Ethereum, is going with hash-based signatures for the validators.
  - aggregation through SNARK proofs (eg. binus, flock, etc)
  - each slot a different XMSS leaf
  - the most conservative choice in terms of security
- hash-based simpler to implement, less potential errors.  
Whereas lattice-based have more subtleties
- hash-based schemes don't have much algebraic structure
  - good for less attacks, not so fun to build stuff with
  - eg. elliptic curves have algebraic structure, allows for signature aggregation, thresholds, etc
- lattice-based schemes have smaller proofs, signatures and public keys
  - better suited for end-user wallets

# Post Quantum Cryptography - Status of the ecosystem

- isogenesis based
- hash based
- codes based
- lattice based
- multivariate based

TODO

# Preliminaries

Recap of concepts:

- Rings
  - ideals
- Vector Spaces
- Modules
- Kernel

# Rings

## Definition (Ring)

A set  $A$  with operations called *addition* and *multiplication* which satisfy the following axioms:

- i.  $A$  with addition alone is an abelian group.
- ii. Multiplication is associative.
- iii. Multiplication is distributive over addition. That is,  $\forall a, b, c \in A$ ,

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$I \subset R$  ( $R$  ring) such that  $0 \in I$  and  $\forall x \in I, r \in R, xr, rx \in I$ .

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Eg: Ring  $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$ , Ideal  $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$  for fixed  $n \in \mathbb{Z}$ .

# Vector spaces

## Definition (Vector space)

A *vector space* over a field  $F$  is a set  $V$ , with two operations  $+$ ,  $\cdot$ , called *vector addition* and *scalar multiplication*, such that

- $V$  with vector addition is an abelian group
- $\forall k \in F$  and  $\vec{a} \in V$ , the scalar product  $k\vec{a}$  is an element of  $V$ , subject to the following conditions:  $\forall k, l \in F, \vec{a}, \vec{b} \in V$ 
  - i.  $k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}$
  - ii.  $(k + l)\vec{a} = k\vec{a} + l\vec{a}$
  - iii.  $k(l\vec{a}) = (kl)\vec{a}$
  - iv.  $1\vec{a} = \vec{a}$
- *basis*: set of vectors that *span* the vector space

# Modules

## Definition (Module)

Let  $A$  be a ring. An  $A$ -module is an Abelian group  $M$  with a multiplication map

$$\begin{aligned} A \times M &\longrightarrow M \\ (f, m) &\longmapsto fm \end{aligned}$$

satisfying  $\forall f, g \in A, \quad m, n \in M$ .

- i.  $f(m \pm n) = fm \pm fn$
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Remark: Vector spaces are defined over fields, a Module is the generalization of a vector space where the scalars come from a ring instead of a field.

# Kernel

## TODO

Intuition: set of inputs that a function completely "erases", ie. they all get mapped to zero.

# Lattices

## Definition (Lattice)

A lattice  $\mathcal{L}$  in  $\mathbb{R}^n$  is the set of all integer linear combinations of  $m$  linearly independent vectors  $b_1, \dots, b_m \in \mathbb{R}^n$ , with  $m \leq n$ .

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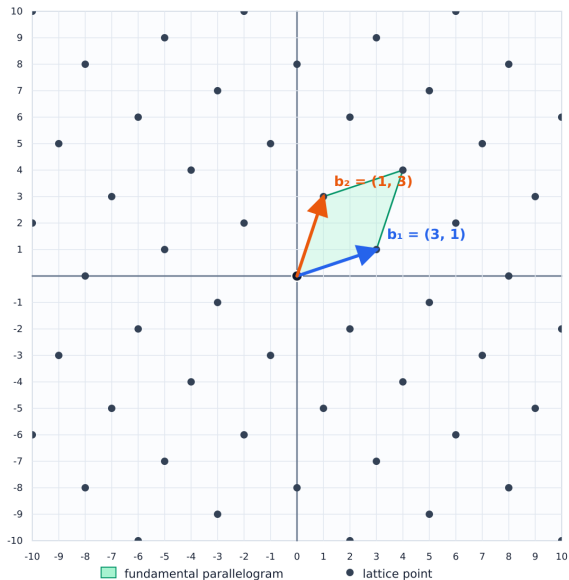
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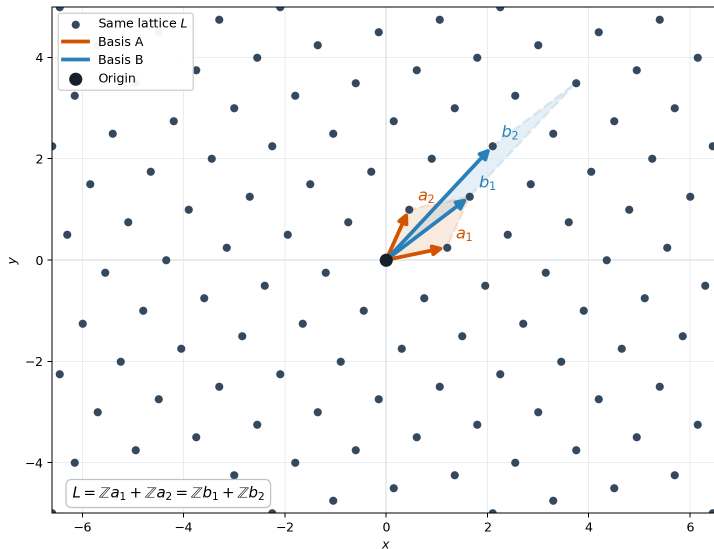
In most cryptographic applications,  $b_1, \dots, b_n \in \mathbb{Z}^n$ , ie.  $\mathcal{L}$  is an integer lattice.

## 2-dimensional lattice

$$\Lambda = \{n b_1 + m b_2 : n, m \in \mathbb{Z}\}, b_1 = (3, 1), b_2 = (1, 3)$$



### One lattice, two different bases



# Lattice properties

a  $n$ -dimensional lattice  $\mathcal{L}$  is

- i. *additive subgroup* of  $\mathbb{R}^n$ :  $\mathcal{L}$  is the nonempty subset of  $\mathbb{R}^n$  closed under vector addition (and subtraction),

$$\text{ie. } \forall x, y \in \mathcal{L}, -x, x + y \in \mathcal{L}$$

- ii. *discrete*: every  $x \in \mathcal{L}$  has a neighborhood in  $\mathbb{R}^n$  in which  $x$  is the only lattice point.

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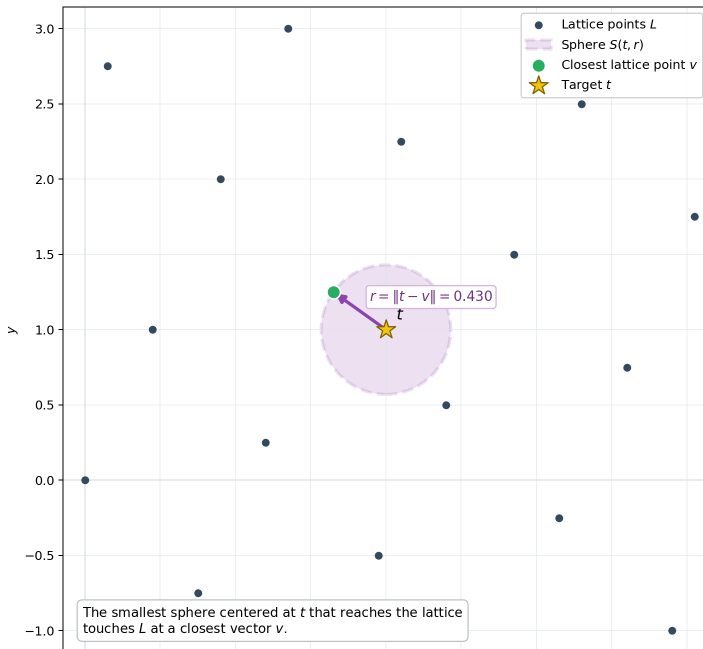
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*Successive minima*  $\lambda_i(\mathcal{L})$ : smallest radius  $r$  s. th.  $\mathcal{L}$  the  $n$ -dimensional ball of radius  $r$  contains  $i$  linearly independent lattice points.

$$\lambda_1(\mathcal{L}) \leq \lambda_2(\mathcal{L}) \leq \dots \leq \lambda_n(\mathcal{L})$$

### Closest Vector Problem: the target-centered sphere



# Lattice properties

recall, a lattice:  $\mathcal{L}(B) := B \cdot \mathbb{Z}^n = \left\{ \sum_{i=1}^n z_i b_i \mid z_i \in \mathbb{Z}^n \right\}$

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- the matrix-vector product  $Bz$  maps an integer coordinate vector  $z \in \mathbb{Z}^n$  into a lattice point in  $\mathbb{Z}^n$ .

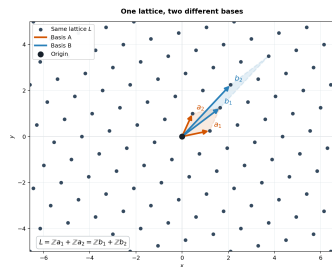
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    - the basis  $B$  is not unique
  - $B \in \mathbb{Z}^{n \times n}$ ,  $n$  vectors, each with  $n$  coordinates  $\implies$  requires  $\mathcal{O}(n^2)$  integers.
    - only if we could have some better lattice structure to represent the lattice with fewer entries...



# q-ary lattice

a full-rank lattice is q-ary if  $q\mathbb{Z}^d \subseteq \mathcal{L} \subseteq \mathbb{Z}^d$

$\implies$  q-ary lattice is an infinite set of integer points, but membership depends only on the residue class modulo  $q$

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q-ary lattice:

$$\mathcal{L}_q^\perp(A) = \ker(\psi_A) = \{z \in \mathbb{Z}^m \mid Az = 0 \pmod{q}\}$$

contains all vectors that are orthogonal modulo  $q$  to the rows of  $A$ .  
The q-ary lattice generated by the rows of  $A$ :

$$\mathcal{L}_q(A) = \{y \in \mathbb{Z}^m \mid y = A^T z \pmod{q} \text{ for some } z \in \mathbb{Z}_q^n\}$$

# Polynomial rings

- The NTRU paper (1996) introduced the usage of polynomial rings, which latter was ported over Ajtai work and other lattices, now is the general approach in Ring-SIS, Ring-LWE, Module-LWE, etc.
- Polynomial rings: represent an  $n$ -dimensional vector  $(a_0, \dots, a_{n-1}) \in \mathbb{Z}_q^n$  as a single ring element (polynomial)

$$a(X) = a_0 + a_1X + \dots + a_{n-1}X^{n-1} \in \mathbb{Z}_q[X]/(X^n + 1)$$

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- Addition:  $a(x) + b(x) = \sum_{i=1}^d (a_i + b_i)X^i$
- Multiplication:

$$a(X) \cdot b(X) = \left( \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} a_i b_j X^{i+j} \right) \in \mathbb{Z}_q[X]/(X^n + 1)$$

Since  $X^n = -1$  then  $X^{n+k} = -X^k$   
 $\implies \mathcal{O}(n^2)$  naively,  $\mathcal{O}(n \log n)$  using NTT.

# Incomplete Timeline

- 1978 Knapsack cryptosystems attacked using lattice methods
- 1996 NTRU encryption
- 2005 Learning With Errors (LWE)
- 2008 Practical secure lattice-signature schemes
- 2009 Fully Homomorphic Encryption (FHE)
- 2016 NewHope (Ring-LWE KEM)
- 2017 Kyber (Module-LWE based) Dilithium (Module-LWE Module-SIS based)
- 2020s standardization: ML-KEM (Kyber) and ML-DSA (Dilithium)

# Hard problems in lattices - SVP and CVP

- In elliptic curve cryptography, we rely on the hardness of the Discrete Logarithm problem:  
given publicly known  $g, p \in \mathbb{G}$ , find privately known  $x \in \mathbb{Z}$  such that  $p = g^x$ .
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find a shortest non-zero vector  $x \in \mathcal{L}$ , ie. that minimizes the Euclidean norm  $\|x\|$ , that is  $\lambda_1(\mathcal{L}) = \|x\|$ .

Intuition: find the shortest possible jump from the origin to another dot

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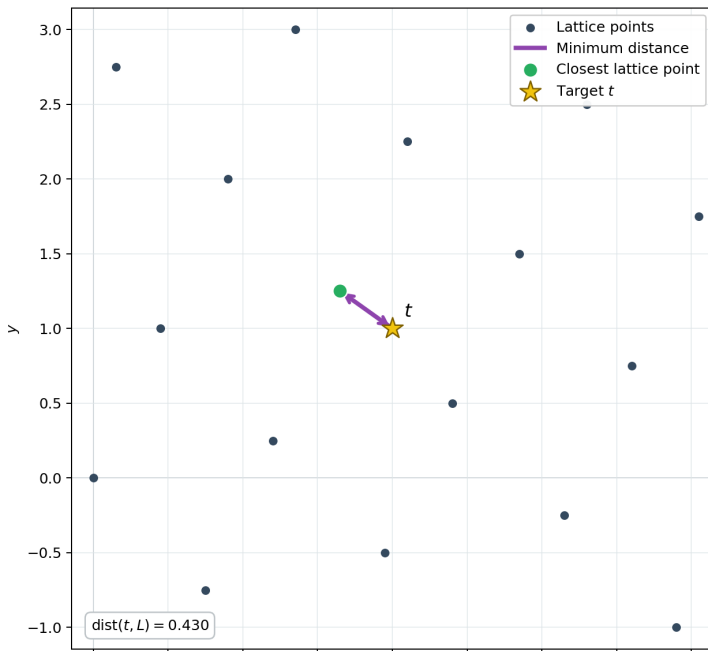
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## Definition (CVP (Closest Vector Problem))

given a vector  $y \in \mathbb{R}^n$  with  $y \notin \mathcal{L}$ , find a nonzero vector  $x \in \mathcal{L}$  that is closest to  $y$ , ie. that minimizes the Euclidean norm  $\|y - x\|$ .

Intuition: find the grid dot that is closest to a given point (off the grid)

# Closest Vector Problem (CVP)



# Why Lattices?

- LLL algorithm is used to solve SVP and CVP.
  - LLL (Lenstra–Lenstra–Lovász): polynomial time method used for lattice basis reduction, which transforms a given basis of a lattice into a new basis that is shorter and nearly orthogonal, which makes SVP and CVP easier to approximate or solve using subsequent algorithms.

Intuition: make the lattice basis less skewed. Same grid of points, just a simpler basis. We don't change the lattice, we change the coordinate system used to describe it. → easier to solve SVP and CVP.

- improvement: BKZ reduction algorithm
- Both SVP and CVP become computationally difficult (NP hard) as the lattice dimension  $n$  grows (even on quantum computers)

# NTRU

Definition (NTRU key recovery problem)

Given  $h(x)$ , find ternary polynomials  $f, g \in \mathcal{T}$  satisfying

$$f(x) \cdot h(x) = g(x) \pmod{q} \in R_q$$

Where  $R_q = \mathbb{Z}_q[X]/(X^n - 1)$ .

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- 1998, was first scheme using polynomial rings, which latter where approached as algebraically structured lattices
  - "lattice-based" bcs recovering the small secret polynomials can be formulated and attacked as a lattice problem
  - earlier lattice usage: in 1978, Knapsack cryptosystems targeted by lattice attacks
- following the approach, in 2002, Micciano modified Ajtai's function to work over polynomial rings getting efficiency improvements

# Hard problems in lattices - SIS

Definition (SIS (Short Integer Solution))

given  $q \in \mathbb{Z}$  and matrix  $A \in \mathbb{Z}_q^{m \times n}$ , find a non-zero short  $z \in \mathbb{Z}^n$  s.th.

$$A \cdot z = 0 \pmod{q}$$

"short":  $z \in \{B, B\}^n$  for some bound  $B$ , often  $z \in \{-1, 0, 1\}^n$ .

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The set of all solutions to the SIS problem forms the ( $q$ -ary) lattice

$$\Lambda_q^\perp(A) = \{z \in \mathbb{Z}^n \mid A \cdot z = 0 \pmod{q}\}$$

where the SIS problem becomes the SVP for  $\Lambda_q^\perp(A)$

Intuition: find a short nonzero vector in the lattice  $\Lambda_q^\perp(A)$

## Definition (Ring SIS (Ring Short Integer Solution))

given a uniformly random matrix  $A \in {}^R R_q^n$ , with elements  $a_i \in R_q$ ,  
find a non-zero integer vector  $z \in R^n$  of norm  $\|z\| \leq \beta$  such that

$$Az = \sum_i a_i \cdot z_i = 0 \in R_q$$

## Definition (Ring SIS (Ring Short Integer Solution))

given a uniformly random matrix  $A \in {}^R R_q^n$ , with elements  $a_i \in R_q$ ,  
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## Definition (MSIS (Module Short Integer Solution))

given a uniformly random matrix  $A \in {}^R R_q^{m \times n}$ , with columns  
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Notice that Module-SIS is a Ring-SIS with  $rank = 1$  ( $m = 1$ ).

- without  $\|z\| \leq \beta$ , it is easy to find a solution via *Gaussian elimination*.
- $\beta < q$ , because otherwise  $z = (q, 0, \dots, 0) \in R^m$  would always be a valid solution.
- $\beta$  and  $m$  must be large enough so that a solution is guaranteed to exist, usually  $\beta \geq \sqrt{\hat{m}}$  and  $m \geq \hat{m}$  where  $\hat{m}$  is the smallest integer greater than  $n \log q$ .

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Again: overall when doing lattice-based cryptography, we will be interested into keeping the values *short*, aka. of small norm.

(example of decomposing values to keep small norms when doing operations in cryptographic protocols (eg. in LaBRADOR))

# Hard problems in lattices

## Definition (LWE (Learning With Errors))

choose a matrix  $A \in {}^R \mathbb{Z}_q^{m \times n}$ , a secret vector  $s \in {}^R \mathbb{Z}_q^n$ , and an error vector  $e \in {}^R [-B, B]^m$ . Let  $b = As + e \pmod{q} \in \mathbb{Z}_q^m$ . Given the pair  $(A, b)$ , the goal is to recover  $s$ .

Search-LWE & Decisional-LWE problems.

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For LWE samples, vector  $b$  is relatively close to exactly one vector in the LWE lattice:

$$\begin{aligned}\Lambda_q(A) &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

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Observe: if  $e = 0$ , is a system of linear equations, which can be efficiently determined using Gaussian elimination over  $\mathbb{Z}_q$ . Hardness comes from the noise/error term  $e \neq 0$  (and the norm bound). Similarly as with SIS, we also have Ring-LWE and Module-LWE variants of LWE.

# Plain - Module - Ring

$$R_q = \mathbb{Z}_q[x]/(x^n + 1)$$

| Assumption | Public linear relation                           | Underlying algebra                                                           |
|------------|--------------------------------------------------|------------------------------------------------------------------------------|
| Plain LWE  | $\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}$ | $A \in \mathbb{Z}_q^{m \times n}$                                            |
| Module-LWE | $\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}$ | $A \in R_q^{k \times k}, \quad \mathbf{s}, \mathbf{e}, \mathbf{b} \in R_q^k$ |
| Ring-LWE   | $b = as + e \pmod{q}$                            | $a, s, e, b \in R_q$                                                         |

| Assumption | Public linear relation                                                                                    | Underlying algebra                                                   |
|------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Plain SIS  | $A\mathbf{z} = \mathbf{0} \pmod{q}, \quad \mathbf{z} \neq \mathbf{0}, \quad \ \mathbf{z}\  \text{ small}$ | $A \in \mathbb{Z}_q^{m \times n}, \quad \mathbf{z} \in \mathbb{Z}^n$ |
| Module-SIS | $A\mathbf{z} = \mathbf{0} \pmod{q}, \quad \mathbf{z} \neq \mathbf{0}, \quad \ \mathbf{z}\  \text{ small}$ | $A \in R_q^{m \times n}, \quad \mathbf{z} \in R^n$                   |
| Ring-SIS   | $az = 0 \pmod{q}, \quad z \neq 0, \quad \ z\  \text{ small}$                                              | $a \in R_q, \quad z \in R$                                           |

# Dual lattices

SIS lattice:

$$\mathcal{L}^\perp(A) = \{z \in \mathbb{Z}^n \mid A \cdot z = 0 \pmod{q}\}$$

LWE lattice:

$$\begin{aligned}\mathcal{L}(A) &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

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Dual of a lattice:

$$\mathcal{L}^* = \{y \in \mathbb{R}^m \mid \langle y, x \rangle \in \mathbb{Z} \forall x \in \mathcal{L}\}$$

ie. all vectors that measure lattice points with integers.

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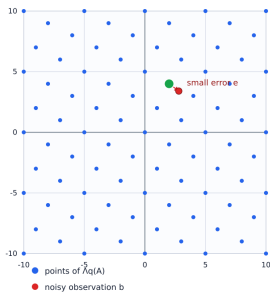
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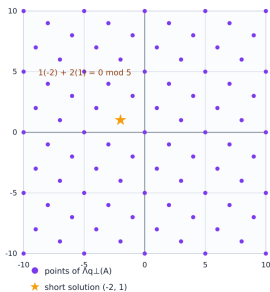
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SIS lattices and LWE lattices are dual to each other

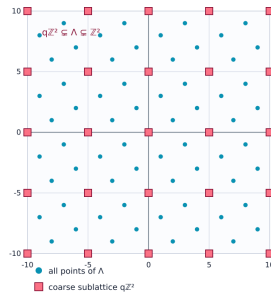
$$\mathcal{L}(A) = q\mathcal{L}^{\perp*}(A) \quad \text{ie.} \quad \frac{1}{q}\mathcal{L}(A) = \mathcal{L}^{\perp*}(A)$$

$q = 5$ Toy LWE: decode near  $\Lambda q(A)$ 

$$\Lambda q(A) = \{s(1,2) + 5z : s \in \mathbb{Z}, z \in \mathbb{Z}^2\}$$

Toy SIS:  $Az = 0 \pmod{q}$ 

$$\Lambda q_{\perp}(A) = \{z \in \mathbb{Z}^2 : ((1,2), z) = 0 \pmod{5}\}$$

q-ary lattice versus  $q\mathbb{Z}^2$ 

$$\Lambda = 5\mathbb{Z}^2 + \mathbb{Z}(2,1) = \{5z + t(2,1) : z \in \mathbb{Z}^2, t \in \mathbb{Z}\}$$

# Ajtai commitments

## Definition (Ajtai commitment)

For a public random matrix  $A \in \mathbb{Z}_q^{m \times n}$ , embedding matrix  $G \in \mathbb{Z}_q^{m \times l}$  and a short random vector  $r \in \{-B, B\}^n$  (for bound  $B$ ), an Ajtai commitment to the message  $x \in \mathbb{Z}_q^l$  (bounded to  $B$ ) is

$$c = Ar + Gx \pmod{q} \in \mathbb{Z}_q^n$$

Observe that this is closely related to the SIS problem ( $Az = 0 \pmod{q}$ ), and it relies on its hardness.

# Lattice-based cryptography

- ML-KEM (Kyber): Key Encapsulation Mechanism. ie. Alice encapsulates a (symmetric) key using Bob's public key
- ML-DSA (Dilithium): signature scheme
- FN-DSA (Falcon): signature scheme, based on NTRU
- LaBRADOR: SNARK scheme
- FHE (BFV, BGV, CKKS, TFHE)

# ML-KEM (Kyber)

- KEM = Key Encapsulation Mechanism
  - Alice encapsulates a (symmetric) key using Bob's public key
- based on Module-LWE

$$R_q = \mathbb{Z}_q[X]/(X^{256} + 1), A \in R_q^{k \times k}$$

**Key generation:**

$$\mathbf{s}, \mathbf{e} \leftarrow \chi^k, \quad \mathbf{t} = A\mathbf{s} + \mathbf{e}$$

$$pk = (A, \mathbf{t}), \quad sk = \mathbf{s}$$

**Encapsulation:**

$$\mathbf{u} = A^T \mathbf{r} + \mathbf{e}_1, \quad v = \mathbf{t}^T \mathbf{r} + e_2 + \text{Encode}(m)$$

$$c = (\mathbf{u}, v), \quad K = \text{KDF}(m, c)$$

**Decapsulation:**

$$m' = \text{Decode}(v - \mathbf{s}^T \mathbf{u})$$

Since

$$v - \mathbf{s}^T \mathbf{u} = \text{Encode}(m) + \mathbf{e}^T \mathbf{r} + e_2 - \mathbf{s}^T \mathbf{e}_1,$$

the message is recovered up to small noise.

# ML-DSA signatures

# Falcon signatures

# LaBRADOR SNARK

# Benchmarks

## Digital signatures

| Scheme           | Assumption              | Public key size | Signature size |
|------------------|-------------------------|-----------------|----------------|
| Falcon-512       | NTRU lattice            | 897 B           | 666 B          |
| ML-DSA-44        | Module-LWE / Module-SIS | 1.31 KB         | 2.42 KB        |
| XMSS-SHA2-10-256 | hash based, stateful    | 64 B            | 2.50 KB        |
| SLH-DSA-128s     | hash based, stateless   | 32 B            | 7.86 KB        |
| ECDSA-secp256k1  | elliptic curves         | 33 B            | 64 B           |

## SNARKs / STARKs

| Scheme              | Assumption / technique        | Proof size  |
|---------------------|-------------------------------|-------------|
| Groth16             | pairings / elliptic curves    | 128 B       |
| LaBRADOR            | Module-SIS                    | 50–60 KB    |
| Flock               | binary fields + coding theory | 200–400 KiB |
| Binius64            | binary fields + sumcheck      | 200–500 KiB |
| Plonky2             | Plonk + FRI                   | 40–200 KB   |
| Plonky3 / FRI-STARK | polynomial IOP + FRI          | 1–3 MiB     |

# Conclusions

- not a single assumption: NTRU, SIS, R-SIS, M-SIS, LWE, R-LWE, M-LWE, ...